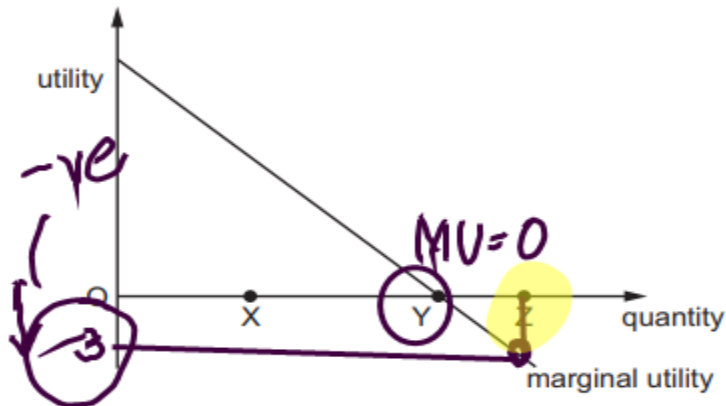


1-C:

The marginal utility diagram shows the relationship between the quantity of a good consumed and the additional utility (satisfaction) gained from each additional unit consumed.



- At Y, marginal utility is zero, meaning that the consumer derives no additional utility from consuming one more unit. This represents that total utility is maximized at **unit Y**.
- At Z, marginal utility is negative, which represents that there will be dis-utility, or in other words, **total utility will fall**.
- Total utility decreases as **quantity consumed increases from Y to Z. To get a better understanding about this, the table given below can be referred to.**

Units	TU	MU
1	10	10
2	20	10
3	28	8
4	33	5
5	35	2
6 <i>Y</i>	35	0
7 <i>Z</i>	34	-1

The diagram also shows that:

- Total utility increases as quantity consumed increases from X to Y (utility peaks at Y).

This explains why key C will be correct.

2-A:

A Pareto improvement is a reallocation of goods that makes at least one consumer better off without making any other consumer worse off.

A)

- Consumer 1 still receives 2 goods (no change)
- Consumer 2 still receives 5 goods (no change)
- Consumer 3 receives 11 goods (an increase from 10)

Consumer 3 benefits from the reallocation and no consumer is worse off., making this a Pareto improvement.

B)

- Consumer 1 receives 4 goods (an increase from 2)
- Consumer 2 still receives 5 goods (no change)
- Consumer 3 receives 9 goods (a decrease from 10)

Consumer 3 is worse off, receiving one less good than initially. This is not a Pareto improvement.

C)

- Consumer 1 receives 4 goods (an increase from 2)
- Consumer 2 receives 4 goods (a decrease from 5)
- Consumer 3 still receives 10 goods (no change)

Consumer 2 is worse off, receiving one less good than initially. This is not a Pareto improvement.

D)

- Consumer 1 receives 6 goods (an increase from 2)
- Consumer 2 receives 6 goods (an increase from 5)
- Consumer 3 receives 6 goods (a decrease from 10)

Consumer 3 is worse off, receiving 4 less goods than initially. This is not a Pareto improvement.

Only option A) meets the criteria for a Pareto improvement, as no consumer is worse off when making someone better off.

3-B:

In indifference theory, it is assumed that the consumer is “rational” which means that the consumer's objective is to maximize their satisfaction or utility, and this objective remains constant. **This means that the consumer will operate at the consumer equilibrium.**

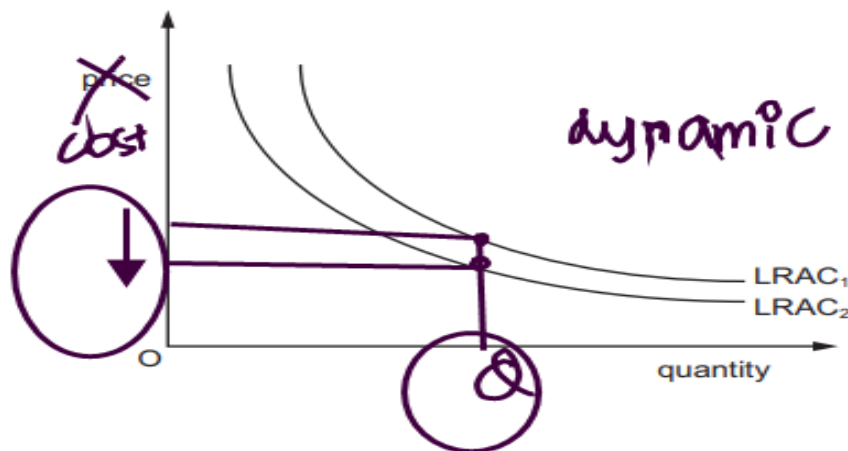
The other options are all assumptions that can change in an indifference theory diagram:

- A) The consumer's income may change, which can lead to a shift in the budget line parallelly, either to the left or the right.
- C) The consumer's tastes and preferences can change, which can lead to a change in the indifference curve (preference curve).
- D) The relative prices of products can change, which can lead to a tilt in the budget line.

4-B:

The shift from LRAC₁ to LRAC₂ indeed represents dynamic efficiency. Dynamic efficiency occurs when a firm or industry is able to reduce its costs and improve its efficiency over time, often through innovation, technological progress by re-investing its super-normal profits.

In this case, the shift from LRAC₁ to LRAC₂ indicates that the firm has achieved dynamic efficiency by reducing its long-run average cost over time, allowing it to produce the same quantity of output at a lower cost.



A) Allocative efficiency refers to the optimal allocation of resources where resources are allocated to produce the goods and services that maximize overall satisfaction. This occurs when the firm produces where $MC=AR$.

C) Pareto efficiency is a concept that refers to an optimal allocation of resources in which no individual can be made better off without making someone else worse off. This means that we are operating on the PPC.

D) Productive efficiency specifically refers to the optimal use of resources to maximize output, given a certain level of technology. This occurs when ATC is minimized which is when $MC=ATC$.



5-D:

- A) $AC = AR$ represents normal profits, where the average cost equals average revenue.
- B) $MC = AR$ represents allocative efficiency, where the marginal cost equals average revenue, indicating optimal resource allocation.
- C) $MC = MR$ represents profit maximization, where the marginal cost equals marginal revenue, indicating the optimal quantity to produce.
- D) $MR = 0$ represents revenue maximization, where the marginal revenue equals zero, indicating the maximum revenue point.

6-B:

Short Run (SR):

- Firms aim to cover Variable Costs (VC) with Total Revenue (TR)
- If $TR \geq VC$, the firm will continue to operate in the SR
- If $TR < VC$, the firm will shut down in the SR

Long Run (LR):

- Firms aim to cover Total Costs (TC) with Total Revenue (TR)
- If $TR \geq TC$, the firm will continue to operate in the LR
- If $TR < TC$, the firm will shut down in the LR

Firm W:

- SR: $TR (\$15) < VC (\$20)$, so it does not operate in the short-run.
- LR: $TR (\$15) < TC$ i.e. $\$30(20+10)$, so it does not operate in the long-run.

Firm X:

- SR: $TR = \$80 > VC = \40 , so it continues to operate in the short run.
- LR: $TR = \$80 < TC = \$90(40+50)$, so it will shut down.

Firm Y:

- SR: $TR = \$500 > VC = \300 , so it continues to operate
- LR: $TR = \$500 < TC = \$600(300+300)$, so it will shut down.

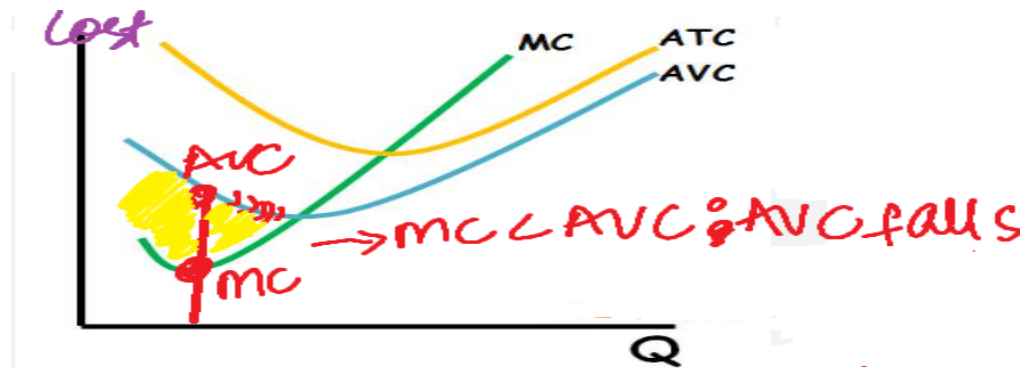
Firm Z:

- SR: $TR = \$550 < VC = \600 , so it does not operate.
- LR: $TR = \$550 < TC$ i.e. $700(600+100)$, so it will not operate in the long-run.

Therefore, in the long run, firms X and Y will shut down because they are not covering their total costs with their total revenue

7-B:

When the short-run marginal is less than the short-run average variable cost it means that the AVC is falling. This is because the marginal cost is the additional cost of producing one more unit, and if it's less than the average variable cost, it means that the average variable cost is decreasing. **This explains why key B is correct.**



A is incorrect because AVC is not constant. Similarly, AVC is lesser than ATC making B incorrect. Additionally, AVC rises when $MC > AVC$, which is not the case here, making C incorrect.

8-A:

When Steel producer X takes over its main competitor, steel producer Y, it means that X has increased its market share and reduced competition. As a result:

- Price elasticity of demand is likely to decrease (become less elastic) because X now has more market power and can influence prices more easily.
- Long-run average cost is likely to decrease due to the elimination of redundant costs, better utilization of resources, and potential economies of scale.

9-D:

External costs, also known as **negative externalities**, are costs that fall on people or entities outside of the firm or market transaction. In this case, road congestion costs are an example of an external cost because they are borne by other road users, rather than by the firm or individual causing the congestion.

A) Distribution costs are an internal cost, as they are borne by the firm itself.

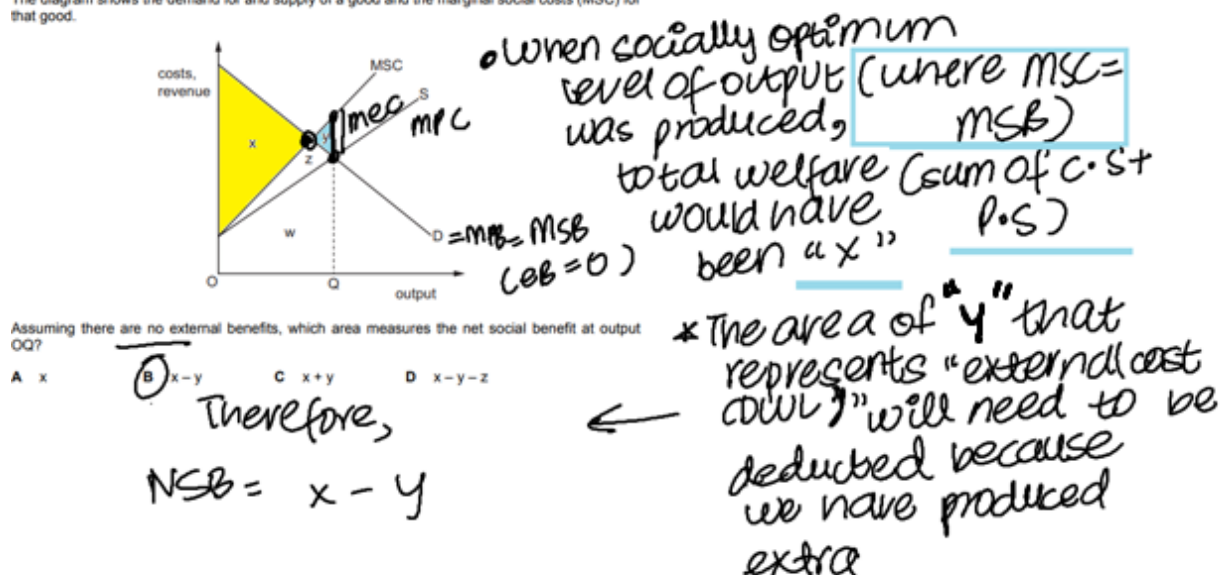
B) Energy costs are also an internal cost, as they are a direct expense for the firm.

C) Labor costs are an internal cost, as they are a direct expense for the firm.

Note: **Internal costs represent "private costs"**.

10-B:

The diagram shows the demand for and supply of a good and the marginal social costs (MSC) for that good.



11-B:

The divorce of ownership and control, also known as the separation of ownership and management, occurs when the owners of a firm (shareholders) are not the same as the managers (agents) who control the firm's operations. This separation can lead to the principal-agent problem, where the agents (managers) may prioritize their own interests over the interests of the principals (owners). **This explains why key B will be correct**

- **A)** The opposite is true; the separation can lead to increased moral hazard. Moral hazard (managers taking risks that benefit themselves but not the owners)
- C)** Productive efficiency is not a direct result of divorce of ownership and control.
- D)** The principal agent problem is most likely to result in the objective of sales revenue maximization being paramount as managerial salaries and bonuses are linked to revenues and not profits.

12-A:

Public goods are characterized by:

- Non-rivalry (consumption by one person does not reduce availability for others)
- Non-excludability (difficult to exclude people from consuming the good because prices cannot be attached due to the free-rider problem.)

The market fails to provide public goods due to the free rider problem because individuals may not pay for something they can consume for free. To address this market failure, the government can directly provide the public good, ensuring it is available to all. **This explains why key A will be correct.**

Here's why the other options are not the best choices:

B) Provision of detailed information to consumers will help to solve the market failure associated with under-consumption of merit goods and over-consumption of demerit goods.

C) Subsidizing firms will not help to solve this market failure as private firms will still not produce public goods because of the free rider problem.

D) Adoption of maximum price controls: This is not relevant to public goods, as the issue is not price but rather the lack of provision due to the free-rider problem.

13-D:

Key take-away: A regulation involves a rule/ law/ limit or a ban.

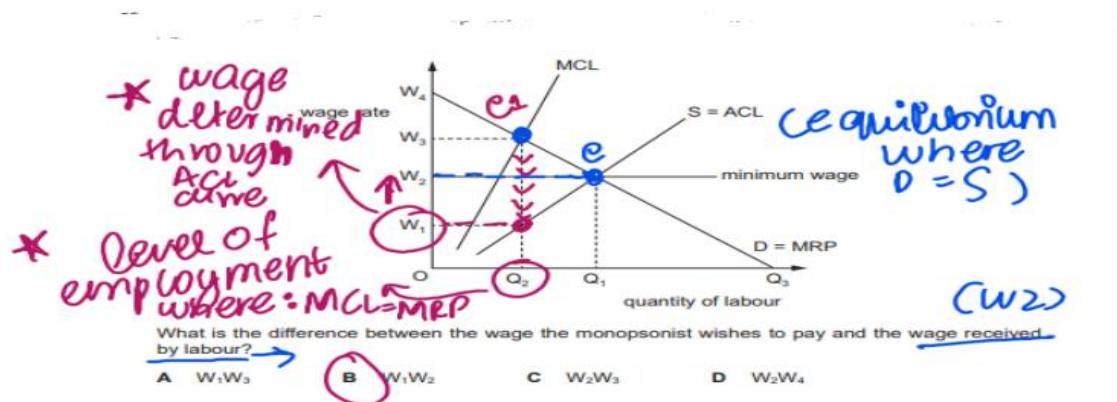
The governor's order is an example of a **regulatory policy**, which is a government directive that mandates specific actions or outcomes for industries, businesses, or individuals. In this case, the regulation bans the sale of new petrol or diesel-powered vehicles in California starting in 2035, effectively promoting the adoption of electric or alternative fuel vehicles.

A) A nudge: A nudge is a subtle policy intervention that influences behavior without mandating a specific action. The ban on petrol and diesel vehicles is not a nudge.

B) A permit to pollute: This option is related to cap-and-trade systems, where companies are allowed to emit pollutants up to a certain limit. The policy in question is not related to pollution permits.

C) Nationalization: Nationalization refers to the government taking ownership or control of industries or assets. The policy in question is a regulation, not a nationalization.

14-B:



This is why, key B, becomes correct.

15-D:

Key take-away:- The “Marginal revenue productivity curve” is equal to the “demand for labor curve”, because the firm takes hiring decisions on the basis of “the contribution of each worker to the output” and to the “revenue”. This is why, it is correct to say that the demand for labor curve is derived from the “marginal revenue product of labor curve”

The MRP of labor ($MPP \times MR$) refers to the additional revenue generated by hiring one more unit of labor. It represents the contribution of that worker to the firm's total revenue. As a firm hires more workers, the MRP reflects the extra output produced and the associated revenue.

Therefore, only option D can be correct because as explained above, MRP is obtained by multiplying the marginal physical product by the marginal revenue.

16-C:

Key take-away: The further the Lorenz curve is from the line of perfect equality, the greater will be the level of inequalities.

C) The Lorenz curves show that Norway has a more equal income distribution (closer to the line of perfect equality) compared to India (further away from the line of perfect equality). This suggests that Norway has a more progressive tax system and a stronger public sector, including public sector pensions, which contributes to greater income equality.

A) This is incorrect because India is more likely to have **higher indirect taxes (that are regressive in nature, because of which, inequalities will be likely to be exacerbated), and not a public sector education system than in Norway. This is because, India has a relatively higher level of inequality than Norway.**

B) This is incorrect because considering that India has higher inequalities, it is more likely to **not have a progressive tax system, and also, there is less likely to be a system of means tested benefits** than in Norway.

D) This is incorrect because considering that Norway has a relatively lesser level of inequality, it is likely to have a lesser percentage of tax generated from indirect taxes. This is because, indirect taxes are regressive in nature, which exacerbate inequalities.

17-B: Note:- **Government failure can occur due to various reasons, including bureaucratic inefficiencies, political pressures, and limited resources.**

Government failure refers to situations where government intervention in the economy leads to inefficient or unintended outcomes.

Option B is actually a reason for government intervention, not failure. **Governments aim to reduce negative externalities, such as pollution or negative health effects, through regulations or policies. This explains why key B is correct.**

A) Out-of-date information: Governments may rely on outdated data, leading to ineffective policies.

C) Difficulty in judging market failure: Governments may struggle to accurately assess the extent of market failures, leading to inadequate interventions.

D) Inability to respond quickly: Governments may be slow to adapt to changing circumstances, rendering their interventions ineffective or obsolete.

All other options explain possible reasons of government failure.

18-D:

A) Giving more modern fishing boats with larger nets to workers who fish for a living encourages overfishing which subsequently leads to depletion of fish stocks. **This will be unsustainable.**

B) Paying bonus payments for larger fish catches to workers who fish for a living encourages overfishing and harmful fishing practices. **This will be unsustainable.**

C) Setting up a fish cooperative so that workers who fish for a living receive a higher price for the fish that they catch will also encourage over fishing in pursuit of receiving bonus payments. **This explains that it will be unsustainable.**

D) Subsidizing the building of a fish farm in order to encourage workers who fish for a living to switch from sea fishing encourages sustainable aquaculture, reduces pressure on wild fish stocks, and promotes a more controlled fishing practice. **This is why, it will be sustainable.**

Key take-away:- In the context of the fishing industry, sustainability refers to the ability to maintain a healthy and productive fish population over time, while also ensuring the long-term viability of the industry and the livelihoods of those who depend on it.

19-D:

This individual is not characterized as structurally unemployed because they have skills and are simply taking time to choose the right job opportunity. This instead represents **search unemployment** which is categorized as **frictional unemployment**. **This explains why key D will be correct.**

Note: **Structural unemployment occurs due to changes in the structure of the industry that occurs when the pattern of demand and supply changes.**

A) The coal miner is unable to find work due to industry decline and is unwilling to retrain which represents that their occupational immobility will make it difficult for them to switch to another industry , making them structurally unemployed.

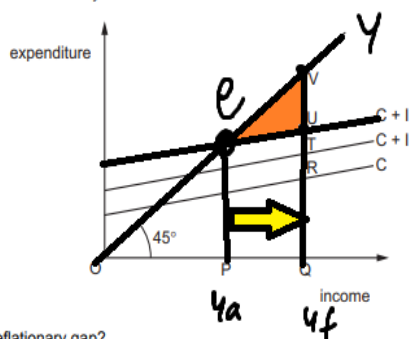
B) The coffee shop worker is out of a job due to a shift in consumer behavior (pandemic-related changes), making it hard to find work in the same field.

This represents a shift in the pattern of demand, which depicts structural unemployment.

C) The individual unable to move with their firm because of geographical immobility due to changing demand is also facing structural unemployment.

20-D:

In the diagram, OP is the equilibrium level of income and OQ is the full employment level of income in a closed economy.



• $P-Q \rightarrow$ negative output gap
 • deflationary gap $\rightarrow VU$

What is the deflationary gap?

- A PQ B RV C TV D UV

$\neg$ output gap $\times$ $\times$

Deflationary gap represents by how much the **aggregate expenditure** falls short of **real income** at the potential level of output that is "Q". This is shown through the distance of "UV". As a result of this, key D will be correct.

21-B:

- Frictional unemployment occurs when workers are temporarily unemployed while searching for a new job or transitioning between jobs. **Improving information about job vacancies can help reduce frictional unemployment since it makes it easier for workers to find jobs that better match their skills and preferences.**

Since it reduces the time taken to search new jobs, this explains that search unemployment which is categorized under frictional unemployment will reduce.

This explains that key B will be correct.

A) Cutting welfare spending on benefits may reduce voluntary unemployment because now the difference between unemployment benefits and the take-home pay of the employed falls.

C) Reducing tax rates may decrease "demand-deficient unemployment" by boosting Aggregate demand as part of expansionary fiscal policy.

D) Raising interest rates (part of contractionary monetary policy) can lead to higher demand-deficient unemployment.

22-C:

In a closed economy with no government sector, an increase in investment spending will generate the largest increase in equilibrium national income when the **value of the investment multiplier** is high.

2 sector multiplier/inv multiplier : $1/(1-mpc)$

- Households have a high marginal propensity to consume (MPC), meaning they spend a large portion of their additional income on consumption.
- **This high MPC ensures that the multiplier effect is large, amplifying the initial increase in investment spending into a larger increase in national income.**
- **This explains why key C will be correct.**

A) Autonomous consumption is unrelated to the multiplier effect and doesn't affect the impact of investment spending on national income.

B) Compulsory saving does reduce consumption, but does not directly affect the impact of investment spending on national income.

D) A high marginal propensity to save (MPS) means households save more, and higher withdrawals represent that the MPC and subsequently the multiplier effect falls, leading to a smaller increase in national income.

23-C:

- Increasing budget deficit (reflationary fiscal policy): The government spends more, injecting money into the economy, stimulating Aggregate demand, and boosting output.
- Decreasing interest rates (monetary policy): Makes borrowing cheaper, encouraging investment, stimulating Aggregate demand as autonomous injections into the circular flow rise which will further boost output. **This explains that key C will be correct.**

24-C:

Inflation: Increase

- Higher oil prices will lead to a rise in import expenditures since the country is a net importer of oil. This will increase costs of production, as a result of which, Aggregate Supply, or the total supply of the economy falls. This will subsequently lead to an **increase in cost push inflation**.

Trade Balance: Worsen

- The country imports more oil than it exports, so higher oil prices increases import expenditures relative to export revenues. This will result in a Balance of trade deficit.

Unemployment: Increase

- When because of a Balance of trade deficit which represents net leakages into the circular flow, Aggregate demand falls, there will be a fall in production, and subsequently demand for labor, which will translate into **demand-deficient unemployment**.

25-A: Note :- **Maximum hours refer to the legal or regulated limit on the number of hours an employee is allowed to work in a given period . This limit is often established to protect workers' health, safety, and well-being, and to ensure they have adequate time for rest and leisure.**

A) This would reduce the effectiveness of a supply-side stimulus because it limits the ability of businesses to respond to increased demand. By imposing maximum hours, businesses may not be able to increase production to meet demand, which would reduce the stimulus's impact on output and employment.

B) This would not reduce the effectiveness of a supply-side stimulus. In fact, an increase in public sector spending would likely complement a supply-side stimulus by boosting Aggregate demand and output.

C) This would not reduce the effectiveness of a supply-side stimulus. New training schemes would enhance workforce skills, increasing productivity and supporting a supply-side stimulus.

D) This would not reduce the effectiveness of a supply-side stimulus. Relaxing import controls would reduce costs and when because of this, Aggregate supply increases, this will further stimulate the economy.

26-D:

Employment: Yes

- The MNC's manufacturing capacity shift will likely create new job opportunities in the developing country.

Investment: Yes

Set-up of a multinational represents an *increase in Foreign Direct Investment*.

- The MNC's decision to shift manufacturing capacity represents a significant investment in the country.

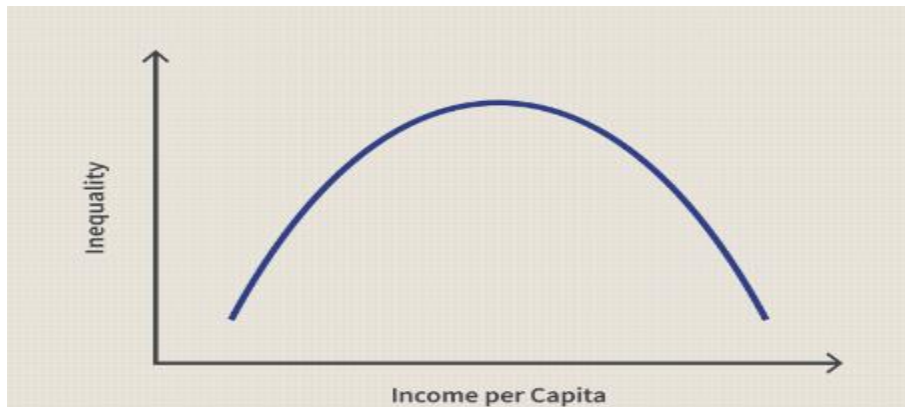
Short-run balance of payments current account: No

- The MNC's investment may lead to an increase in imports of capital goods and technology, which can worsen the current account balance in the **short run**.
- Additionally, the MNC may **repatriate profits** back to its home country, also worsening the current account balance. Repatriation of profits will lead to outflows from the primary income section of the current account.

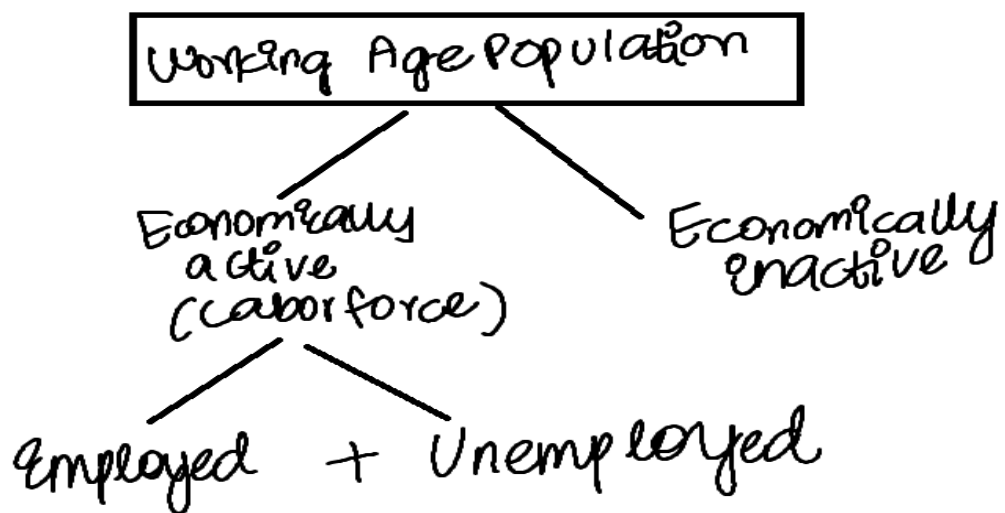
Note: Most importantly, balance of payments current account will improve in the long-run due to an increase in exports done by the multinational.

27-B:

The **Kuznets curve** is a graphical representation of the relationship between economic growth and income inequality. It plots GDP per capita (a measure of economic growth) against the level of **income inequality (variable X)**, often measured by the Gini coefficient. The curve suggests that as an economy grows, income inequality initially increases, but eventually decreases as the economy reaches a certain level of development. **This explains why key B will be correct.**



28-B:



The labor force includes:

- Unemployed: Yes, those actively seeking work are considered part of the labor force.
- **Economically Inactive: No, people who are not working and not seeking work (e.g., stay-at-home parents, retirees) are not considered part of the labor force.**
- Self-employed: Yes, individuals who work for themselves, running their own businesses are considered part of the labor force.

So, the labor force includes both the unemployed and the self-employed, but not the inactive population.

29-B

A) This is incorrect because the less developed regions have the largest urban populations, actual or predicted, in both years.

B) This is true. The table shows that the More Developed regions have a relatively small increase in urbanization compared to the other regions, which have larger increases.

Region	Urbanization Rate (2010)	Urbanization Rate (2050)	Difference
More Developed	75%	86%	+11%
Less Developed	45%	66%	+21%
Least Developed	29%	55%	+26%

C –

Here are the calculations for the population growth rates by 2050:

Population Growth Rates by 2050

- More Developed Regions: $(1275 - 1237) / 1237 = 3.1\%$ increase
- Less Developed Regions: $(7875 - 5671) / 5671 = 38.7\%$ increase
- Least Developed Regions: $(1673 - 855) / 855 = 95.8\%$ increase

This is incorrect because the type of region with the greatest population in 2010 is indeed the Less Developed Region, and the region predicted to have the highest rate of population growth rate is least developed region.

D) The less Developed Regions had the greatest urban population in 2010 (2556 million), but they do not have the highest level of urbanization by 2050. Instead, the More Developed Regions have the highest level of urbanization by 2050 (86%), while the Less Developed Regions have a lower urbanization rate (66%).

30-A:

-> Gini-coefficient is calculated by dividing the region between the line of the perfect equality and the Lorenz curve i.e. X with the total area under the line of perfect equality that is X+Y.

Therefore, the formula of gini-coefficient will be:

$X/(X+Y)$ or X divided by (X plus Y)

